MILITARY SPECIFICATION

ACCELEROMETER, INTEGRATING, M9

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of accelerometer entitled Accelerometer, Integrating, M9 which is used in the Sergeant warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8830423	- Accelerometer, Integrating, M9.
EL-8854156	- Equipment List.

FSC: 1135

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The accelerometer shall comply with all requirements specified on the drawing (dwg) 8830423, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Functional operation. The accelerometer shall have an operational gravity (g)-second product of 8.0 + 1.75 - 2.00 g-seconds when initially tested as specified in 4.4.1. At any time following this initial test, and during any other functional or environmental tests, the operational g-second product shall be 8.0 + 1.75 - 2.75 g-seconds.

3.6 Threshold. The accelerometer shall have a threshold of 2.9 ± 0.3 g when initially tested as specified in 4.4.2. At any time following this initial test, and during any other functional or environmental test, the threshold shall be 2.9 ± 0.6 g.

3.7 Normally open circuit resistance. The normally opened (NO) circuit resistance shall not exceed 0.10 ohms when initially tested as specified in 4.4.3. At any time following this initial test, and during any other functional or environmental test, the NO circuit resistance shall not exceed 0.20 ohm.

3.8 Normally closed circuit resistance. The normally closed (NC) circuit resistance shall not exceed 1.0 ohm when initially tested as specified in 4.4.4. The open circuit resistance shall exceed 1 megohm when tested as specified in 4.4.4.

3.9 Insulation resistance. The insulation resistance of the NO circuit shall be greater than 50 megohms when tested as specified in 4.4.5.

3.10 Case leakage. The leak rate of the component case shall not exceed 5 x 10-6 cubic centimeters (cc) of helium per second when tested as specified in 4.4.6.

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3.11 Environmental testing. The accelerometer shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The accelerometer shall also comply with 4.3.3.1 before and after each environmental test. Testing shall be as specified in 4.4.7. Tests shall be conducted in the order specified.

3.11.1 Function at temperature extremes. Exposure to temperature extremes of from -65 °F. ± 5°F. to 160°F. ± 5°F. shall not cause impairment of operation during or after exposure when tested as specified in 4.4.7.1.

3.11.2 Mechanical shock. The accelerometer shall withstand shocks of 30 ± 5 g three times in each direction along the three mutually perpendicular axes shown in Figure 1 for a total of 18 shocks when tested as specified in 4.4.7.2. During the test no circuit closure greater than 100 microseconds shall occur. After completion of this test, the unit shall meet the requirements of 3.5 through 3.10, except that in the functional operation test (3.5) a single measurement for functioning shall be made at 3.5 ± 0.1 g level and in the NO circuit resistance test (3.7) a single resistance measurement shall be made at 10 ± 1.0 g level.

3.11.3 Transportation vibration. The accelerometer shall withstand vibration for 15 minutes in each of three mutually perpendicular axes, as shown in Figure 1, at the frequencies and amplitudes specified below without evidence of damage or affecting function when tested as specified in 4.4.7.3. The frequency shall be varied from 8 to 500 to 8 Hertz (Hz) in 15 minutes.

Frequency Range (Hz)	Cycle Time	Constant Acceleration or Double Amplitude
8-55-8	7.5 minutes	0.060 ± 0.006 inch
55-500-55	7.5 minutes	10 ± 1 g

3.11.4 Operational vibration. The accelerometer shall withstand without evidence of damage or affecting functioning a programmed sequence of band-limited white Gaussian noise and combined noise and sine wave vibration without evidence of damage or affecting function when tested as specified in 4.4.7.4. The total time of test shall be 210 seconds; 70 seconds in each of the three mutually perpendicular axes shown in Figure 1. The complex vibration test will consist of the following five sections, programmed continuously.

a. White Gaussian noise at a level of 22 gravity units (g) root mean square (RMS) band-limited between 15 and 1500 Hz for one (1) second.

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b. A 4 g RMS swept sine wave superimposed on 7.5 g RMS of white Gaussian noise band-limited between 15 and 1500 Hz for 15 seconds. The sine wave shall be swept from 800 to 1300 Hz at a rate increasing directly with frequency, one such sweep taking the full 15 seconds.
c. An 8 g RMS swept sine wave superimposed on 15 g RMS of white Gaussian noise band-limited between 15 and 1500 Hz for 15 seconds. The sine wave shall be swept from 1300 to 800 Hz at a rate decreasing directly with frequency, one such sweep taking the full 15 seconds.
d. A 4 g RMS sine wave superimposed on 7.5 g RMS of white Gaussian noise band-limited between 15 and 1500 Hz for 38 seconds. The sine wave shall be swept from 15 to 1500 Hz and back to 15 Hz at a rate increasing directly with frequency. One such cycle shall take the full 38 seconds.
e. White Gaussian noise at a level of 22 g RMS band-limited between 15 and 1500 Hz for one (1) second.

Following the vibration test, the accelerometer shall meet the requirements of 3.5 through 3.10 except that in the functional operation test (3.5) a single measurement for functioning shall be made at 3.5 + 0.1 g level and in the NO circuit resistance test (3.7) a single resistance measurement shall be made at 10 ± 1.0 g level.

3.12 Assembly Environment. Final assembly shall be in a clean atmosphere maintained at a relative humidity of 50 percent maximum, a temperature of 72 + 5°F. and a particle count of 25,000 particles one micron or larger per cubic foot maximum as determined by a Bausch and Lomb dust counter or equivalent.

3.13 Soldering. Soldering as applicable shall comply with the requirements specified in MIL-STD-1460.

3.14 Workmanship.

3.14.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.14.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.14.3 Burrs. Parts shall be free from burrs.

3.14.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.14.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8830423) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Functional operation (see 3.5)		X
3. Threshold (see 3.6)		X
4. Normally open circuit resistance (see 3.7)		X
5. Normally closed circuit resistance (see 3.8)		X
6. Insulation resistance (see 3.9)		X
7. Case leakage (see 3.10)		X
8. Function at temperature extremes (see 3.11.1)		X
9. Mechanical shock (see 3.11.2)		X
10. Transportation vibration (see 3.11.3)		X
11. Operational vibration (see 3.11.4)	X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

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4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Accelerometer, Integrating (see dwg 8830423).

Categories	Characteristics	Method of inspection
Critical:		
	1. Presence and correctness of orientation marking	Visual
Major:		
	101. Overall length	Gage
	102. 0.150 maximum fill tube length	Gage
	103. Terminals aligned within 7 degrees	Gage
	104. Objective evidence relative to material missing or improper (see 3.1)	Visual
	105. Evidence of poor workmanship (see 3.14)	Visual
Minor:		
	201. Legibility and completeness of nameplate information	Visual

4.3.2.1.2 Preparation for delivery (see 5).

Categories	Characteristics	Method of inspection
Critical:	None defined.	
Major:		
	101. Preservation and packaging not in accordance with MIL-STD-1169 (see 5.1)	Visual

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102. Packing not in accordance with MIL-STD-1169 (see 5.2).....Visual
103. Marking not in accordance with MIL-STD-1169 (see 5.3).....Visual

Minor:
201. Evidence of poor workmanship (see 3.14).....Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below.

Test	Test Classification
a. Insulation resistance	Major
b. Normally closed circuit resistance	Major
c. Threshold	Major
d. Functional operation	Major
e. Normally open circuit resistance	Major
f. Case leakage	Major
g. Function at temperature extremes	Major
h. Mechanical shock	Major
i. Transportation vibration	Major
j. Operational vibration	Major

Of these tests,j shall be considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Insulation resistance, normally closed circuit resistance, threshold, functional operation, normally open circuit resistance, and case leakage. Each accelerometer shall be tested in accordance with 4.4.1 through 4.4.6. If any item fails to comply with 3.5 through 3.10, the item shall be rejected and removed from the lot.

4.3.3.2 Function at temperature extremes (see 3.11.1). Sample accelerometers shall be selected from each lot in accordance with Table IV and subjected to this test (see 4.4.7.1). Allowable defects shall be as indicated in the column of Table IV headed "C." A defect is a characteristic that exhibits an out-of-specification reading at least once and, when retested five times, repeats the out-of-specification value. The re-test shall be performed at the conditions of the original test. Any sample failing the acceptance criteria of Table IV shall result in 100% screening of the balance of the lot for all portions of the test through the failed portion. Following screening, a new sample shall be drawn from the lot in accordance with Table IV and subjected to any remaining

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portions of the test. Any subsequent defects shall result in continued screening as above. Any unit having more than one out-of-specification value, even though it is not classified defective as defined above, shall be removed from the lot.

Table IV
Lot sampling and acceptance criteria

Lot Size	Sample Size	C
0-110	15	0
111-300	35	1
301-500	50	2

4.3.3.3 Mechanical shock (see 3.11.2). Sample accelerometers shall be selected from the sample units which have successfully completed 4.3.3.2, in accordance with Table V and subjected to this test (see 4.4.7.2). Allowable defects shall be as indicated in column of Table V headed "C." A defect is a characteristic that exhibits an out-of-specification reading at least once and, when re-tested five times, repeats the out-of-specification value. The re-test shall be performed at the conditions of the original test. Any sample failing the acceptance criteria of Table V shall result in 100% screening of the balance of lot all portions of the test through the failed portion. Following screening, a new sample shall be drawn from the lot in accordance with Table V and subjected to any remaining portions of the test. Any subsequent defects shall result in continued screening as above. Any unit having more than out-of-specification value, even though it is not classified defective as defined above, shall be removed from the lot.

Table V

Lot sampling and acceptance criteria

Lot Size	Sample Size	C
0-110	5	0
111-300	7	0
301-500	10	1

4.3.3.4 Transportation vibration (see 3.11.3) followed by operational vibration (see 3.11.4). The sample accelerometers which have successfully completed 4.3.3.3 shall be subjected to these tests (see 4.4.7.3 and 4.4.7.4) in the quantities shown in the column headed "n1" of Table VI. Allowable defects shall be as indicated in the column of Table VI headed "C." A defect is a characteristic that exhibits an out-of-specification reading at least once and, when retested five times, repeats the out-of-specification value. The retest shall be performed at the conditions of the original test. If the number of defects on the first sample (n1) is greater than "C" but less than "R,"

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a second sample shall be selected in accordance with the column of Table VI headed "n2" and subjected to this test. The lot shall be accepted if the number of defects in the combined sample (n1 + n2) is less than "R." If the number of defects in the first sample (n1) or in the combined sample (n1 + n2) is equal to or greater than "R" acceptance of the lot from which the sample(s) was taken shall be suspended until corrective action is taken on the lot or conformance to prescribed criteria is demonstrated. The accelerometers selected in accordance with the column headed "n2" shall be taken from the sample in 4.3.3.2. These units comply with the test in 4.3.3.3 prior to being subjected to this test.

Table VI
Lot sampling and acceptance criteria

Lot Size	Sample Size		C	R
	n1	n2		
0-110	5	10	0	2
111-300	7	14	0	3
301-500	10	20	1	4

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (See EL-8854156). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

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4.4.1 Functional operation. The accelerometer shall be mounted on a centrifuge in a suitable jig. This jig may be capable of holding one or more units securely for concurrent testing. The unit(s) shall be positioned by the jig such that acceleration of the centrifuge will not cause the NC monitor circuit to open. When the acceleration of the component is at the value specified, the orientation of the component shall be changed such that the centrifugal acceleration will be along the longitudinal axis of the component in the direction indicated thereon within ± 5° of arc. The time required to perform the final 90° of this change of orientation shall not exceed 0.050 seconds. Three g-second measurements shall be made at the 3.5 ± 0.1 g level and three g-second measurements shall be made at the 10 ± 1.0 g level. When determining g-second product, the actual acceleration shall be multiplied by the measured time between NC circuit opening and NO circuit closing. The NC circuit shall be closed prior to initiating each test sequence. The tolerance on acceleration is given to aid in setting the centrifuge speed. The radius arm of the centrifuge shall be the distance from the centrifuge center to a point 1.125 inches from the flat surface at the NO end of the component when it is in the position causing actuation as noted above.

4.4.2 Threshold. The unit shall be mounted in the jig used for the functional operation orientated such that the centrifugal force will be along the longitudinal axis of the component within ± 5° of arc in a direction that will tend to move the piston toward the component NO circuit end. The centrifuge shall be brought slowly from rest to that speed which will cause the NC circuit to open. Without increasing centrifuge speed, the NO circuit shall subsequently close.

4.4.3 Normally open circuit resistance. The unit shall be mounted in a sustained acceleration field. With a source potential of 6 VDC maximum, a current of 100 milliampere (ma) maximum shall be passed through the circuit. Resistance shall be computed from the voltage drop across the two NO circuit leads when measured by a high impedance meter. One value shall be taken in a 3.5 ± 0.1 g-field and one value shall be taken in a 10 ± 1.0 g-field.

4.4.4 Normally closed circuit resistance. With the unit in a 1 g maximum acceleration field and with a source potential of 6 VDC maximum, a current of 100 ma maximum shall be passed through the NC circuit. Resistance shall be computed from the voltage drop across the two NC circuit leads when measured by a high impedance meter. NC circuit open resistance shall be measured with the device mounted on a centrifuge and in an acceleration field sufficient to maintain the NC circuit open.

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4.4.5 Insulation resistance. With the NC circuit closed, a test voltage of 500 ± 50 volts limited to 5 MA current shall be applied for a minimum of one second. The potential shall be applied between the NO circuit leads and also between each NO circuit lead and the case. The resistance shall be greater than 50 megohms.

4.4.6 Case leakage. The component shall be tested for leaks with a mass spectrometer.

4.4.7 Environmental testing (see 3.11).

4.4.7.1 Function at temperature extremes. The units shall be subjected to air temperature extremes listed in Table VII.

Table VII

Air Temperature	Soak Time
-65°F.	1 hour
Ambient	1 hour
160°F.	1 hour
Ambient	1 hour
-65°F.	1 hour
Ambient	1 hour
160"F.	1 hour

After completion of the above cycle, the units shall be stabilized at ambient (see 6.4) and tested to the requirements of 3.10. The unit shall then be mounted as specified in paragraph 4.4.1 and subjected to an air temperature of -65°F. for 1 hour. At the end of one hour and while at -65°F., the unit shall be tested to the requirements of 3.5 through 3.8 except that in the functional operational test (3.5) a single measurement for functioning shall be made at 3.5 ± 0.1 g-level and in the NO circuit resistance test (3.7) a single resistance measurement shall be made at 10 ± 1.0 g-level. After completion of this test, the component shall be stabilized at room temperature and then subjected to an air temperature of 160°F. for one hour. At the end of the one hour period and while still at 160°F., the unit shall be tested to the requirements of 3.5 through 3.8 except that a single measurement for functioning shall be made at 3.5 ± 0.1 g-level and in the NO circuit resistance test (3.7) a single resistance measurement shall be made at 10 ± 1.0 g-level. After completion of this test, the component shall be stabilized at room temperature and shall be tested to the requirements of 3.9 and 3.10.

4.4.7.2 Mechanical shock. The unit shall be rigidly mounted in a suitable fixture and subjected to a total of 18 shocks. Three 30 ± 5 g hocks shall be applied in both directions of each mutually perpendicular axis, as shown in Figure 1.

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Duration of shock shall be 11 ± 2 milliseconds (ms) as measured between points which are 10 percent of the peaks. The time to peak shall be 5.5 ± 1 ms. The accelerometer NO circuit leads shall be monitored during this test for circuit closure. After completion of this test, the unit shall be tested to the requirements of 3.5 through 3.10 except that in the functional operation test (3.5) a single measurement for functioning shall be made at the 3.5 ± 0.1 g-level and in the NO circuit resistance test (3.7) a single resistance measurement shall be made at the 10 ± 1.0 g-level.

4.4.7.3 Transportation vibration. The accelerometer shall be mounted on a vibration machine and vibrated for 15 minutes in each of three mutually perpendicular axes, as shown in Figure 1, at the frequencies and amplitudes specified in 3.11.3. Total test time for each initiator shall be 45 minutes.

4.4.7.4 Operational vibration. The accelerometer, after completion of the transportation vibration test, shall be vibrated in three mutually perpendicular axes, as shown in Figure 1, at the frequencies and acceleration specified in 3.11.4 using a standard noise or complex wave tape. The tape shall have the characteristics listed in Table VIII.

Table VIII

Time (Sec)	Type of Signal	Ratio of Test Level to Calibration Level	RMS g Level
0-20	Noise (Calibration) (1)	1	Set gain to give 2 g RMS
20-30	None	-	-
30-31	Noise (1)	11	22
31-46	Noise (1) plus sinusoid (2)	4.25	8.5
	Noise alone	3.75	7.5
	Sinusoid alone	2	4
46-61	Noise (1) plus sinusoid (3)	8.5	17
	Noise alone	7.5	15
	Sinusoid alone	4	8
61-99	Noise (1) plus sinusoid (4)	4.25	8.5
	Noise alone	3.75	7.5
	Sinusoid alone	2	4
99-100	Noise (1)	11	22

(1) Band-limited (15-1500 Hertz (Hz)) white Gaussian noise.

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(2) Sine wave swept from 800 to 1300 Hz at a rate increasing directly with frequency. One full sweep in 15 seconds.
(3) Sine wave swept from 1300 to 800 Hz at a rate decreasing directly with frequency. One full sweep in 15 seconds.
(4) Sine wave swept from 15 to 1500 Hz and back to 15 Hz at a rate increasing directly with frequency. One full cycle in 38 seconds.

For creating the random noise, the output of a General Radio random noise generator, or its equivalent, may be used if proper care is taken to insure the correct amplitude distribution of the signal. For testing the noise generator or any other associated equipment, the use of an amplitude distribution analyzer is suggested. For insuring the correct noise bandwidth, a filter with an asymptatic slope of at least 24 decibel (db) per octave and 3 db points at 15 and 1500 Hz shall be considered acceptable. Following the vibration the initiator shall be tested for the requirements of 3.5 through 3.10 except that in the functional operational test (3.5) a single measurement for functioning shall be made at the 3.5 $\pm$ 0.1 g level and in the NO circuit resistance test (3.7) a single resistance measure shall be made at the 10 $\pm$ 1.0 g level.

4.4.7.4.1 Method and procedure with the standard noise tape. Before applying the tape, the transfer function or frequency response of the entire vibration system shall be as flat as is consistent with the present state of the art. Plus or minus 3 db or better over the band from 15 to 1500 Hz shall be considered as an objective in this respect. After proper equalization of the system, the standard noise tape is applied. The overall gain of the shake system is then adjusted so that with the noise calibration signal, the g level listed in 4.4.7.4 is obtained. With the same gain setting, the tape is run for the remainder of the test and the g levels obtained will conform to the levels listed in the table of 4.4.7.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The accelerometer shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The accelerometer shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

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5.2 Packing.

5.2.1 Levels A and C. The accelerometer shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2067 (Rev. 1) dated 1 September 1965.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No.1135-A178

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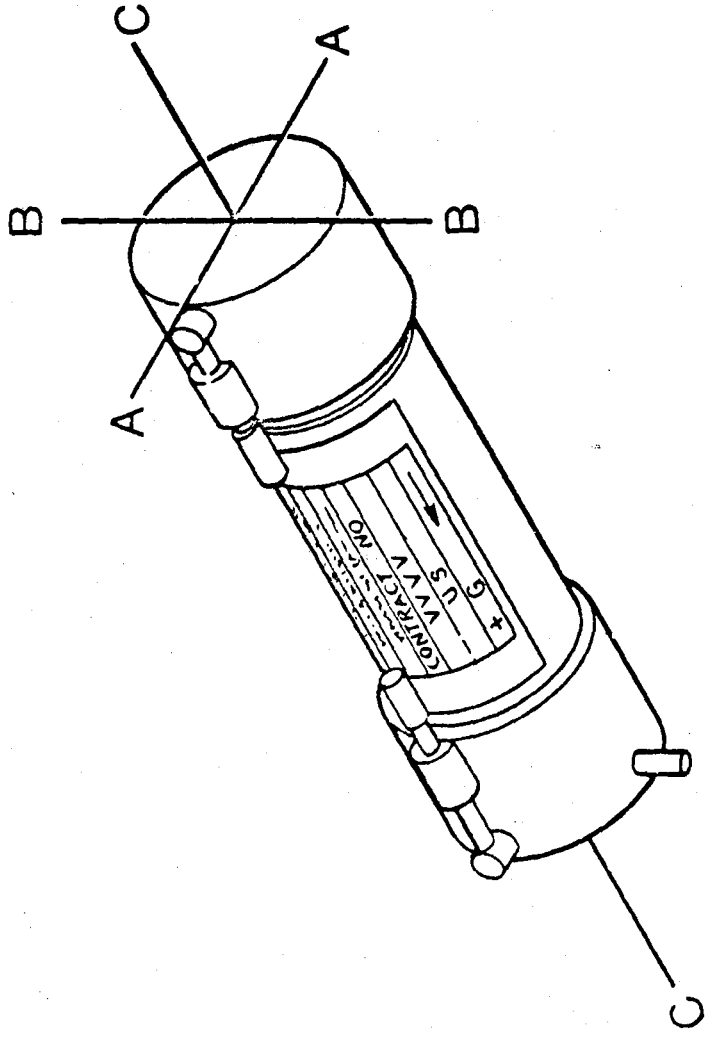


Figure 1 M9 Assembly

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